MBEYA UNIVERSITY OF SCIENCE AND TECHNOLOGY

COLLEGE OF INFOMATION AND COMMUNICATION TECHNOLOGY MODULE

NAME; MOBILE APPLICATION DEVELOPMENT.

MODULE CODE; 8203

FACILITATOR; RUKOIJO A.

MODE OF ASSIGNMENT; GROUP PROJECT

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SEMISTER; ONE.

PROJECT TITLE; ON-DEMAND CLEANING AND SANITATION SERVICES PLATFORM

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SafishaNasi APP

1. Introduction

1.1 Background

With the increasing demand for cleaning services in homes, offices, and institutions, many people face difficulties finding reliable cleaners quickly. Most traditional methods rely on word of mouth, local contacts, or informal arrangements, which are often inefficient and unreliable.

The SafishaNasi App was developed to solve this problem by providing an online platform where customers can easily find and book professional cleaning services. The application connects customers with cleaners and allows bookings to be managed digitally through a simple web interface.

The system also helps cleaners receive job requests and enables administrators to manage the platform effectively.

1.2 Problem Statement

Many households, offices, and schools struggle to find trusted cleaning services efficiently. Traditional systems lack digital booking, centralized service management, and proper communication between customers and cleaners.

The SafishaNasi App addresses this issue by providing a modern web-based platform that simplifies the process of booking cleaning services.

1.3 Project Objectives

- To develop a web-based platform for booking cleaning services.
- To allow customers to easily view available cleaning services.
- To enable customers to book cleaning services online.
- To provide cleaners with a system to receive job requests.
- To allow administrators to manage services and monitor bookings.

2. System Requirements

2.1 Functional Requirements

User Registration – Customers and cleaners must be able to create accounts in the system.

User Login – Users must log into the system using their email and password.

View Services – Customers must be able to view available cleaning services such as:

- Home Cleaning
- Office Cleaning
- School / College Cleaning

Booking Services – Customers must be able to book a cleaning service through the platform.

Cleaner Dashboard – Cleaners must be able to view available cleaning jobs assigned to them.

Admin Management – The admin must be able to manage services and monitor system activities.

CRUD Operations – The system must allow Create, Read, Update, and Delete operations for services and bookings.

2.2 Non Functional Requirements

Usability – The system must be easy to use and understandable for all users.

Performance – The application must respond quickly and handle multiple users.

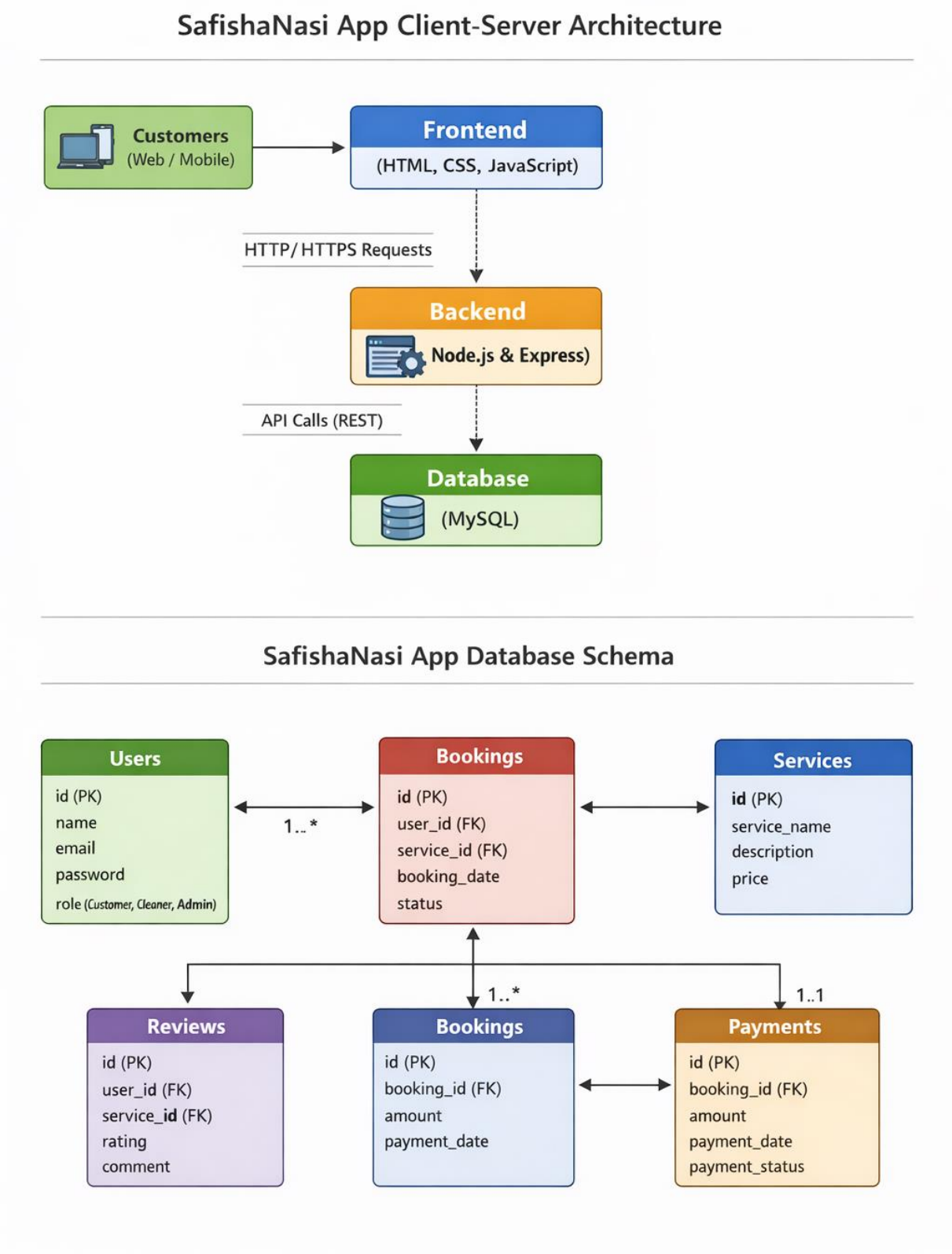
Security – User data must be protected using authentication mechanisms.

Responsiveness – The system must work well on both mobile devices and desktop computers.

Maintainability – The code must be well organized and easy to maintain.

3. System Architecture

The SafishaNasi App follows a client-server architecture consisting of three main components.



Frontend
The frontend displays the user interface and communicates with the backend using API requests.
Technologies used include HTML, CSS, JavaScript, and Font Awesome icons.

Backend
The backend processes requests, handles business logic, and communicates with the database. Technologies used include Node.js and Express.js.

Database
The database stores application data such as users, services, and bookings.
Technology used: MySQL database managed through phpMyAdmin.

4. Project Folder Structure
The project is organized into two main sections: frontend and backend. This structure helps maintain a clean and organized project.

5. Database Design
Users Table – stores user account information (id, name, email, password, role).

Services Table – stores available cleaning services (id, service_name, description, price).

Bookings Table – stores customer bookings (id, user_id, service_id, booking_date, status).

Reviews Table – stores feedback from customers.

Payments Table – stores payment information related to bookings.

6. Security Considerations
Authentication – Users must log in using credentials before accessing system services.

Password Protection – Passwords are stored securely using hashing techniques.

Token-Based Authentication – Tokens maintain secure communication between frontend and backend.

Role-Based Access Control – Different user roles have different permissions (Customer, Cleaner, Admin).

7. Implementation
The SafishaNasi App was implemented using a full stack web development approach.

Frontend Implementation – HTML, CSS, and JavaScript were used to build interactive user interfaces.

Backend Implementation – Node.js and Express.js were used to build API endpoints.

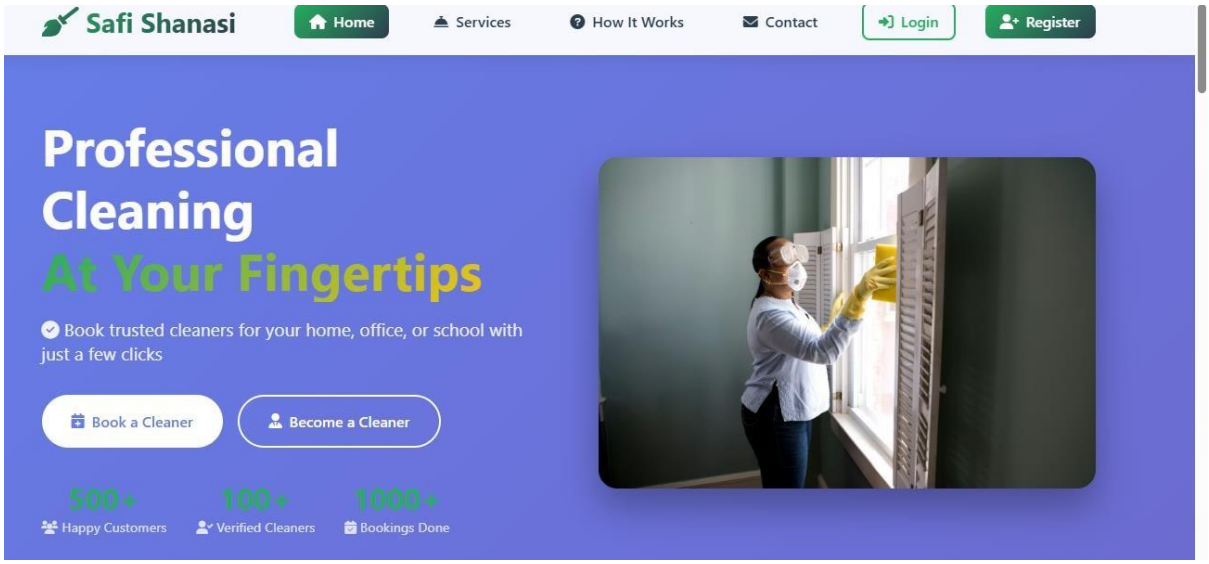
Database Implementation – MySQL stores application data and connects with the backend.

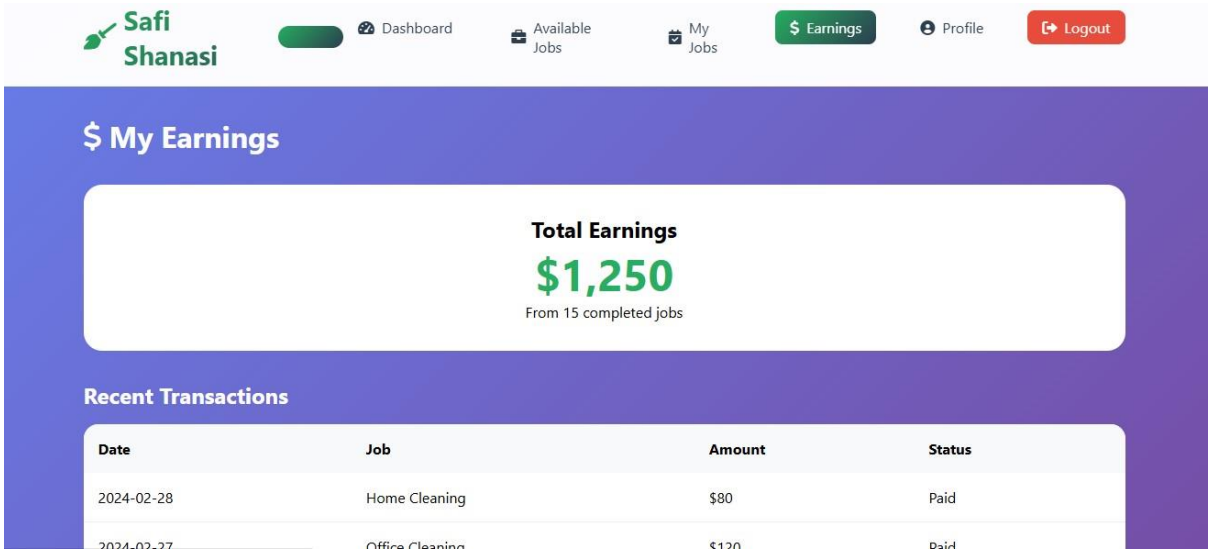
8. Testing and Results

The system was tested to ensure all major features work correctly including:

- User registration
- User login
- Viewing services
- Booking services
- Database connectivity
- Dashboard functionality

Screenshots of the system are included in this section showing different parts of the system we explained





Create Account

Customer | Cleaner | Admin

Full Name

Email address

Phone Number (optional)

Password (min. 6 characters)

Register

Already have an account? Login

Safi Shanasi

Dashboard | Available Jobs | My Jobs | Earnings | Profile | Logout

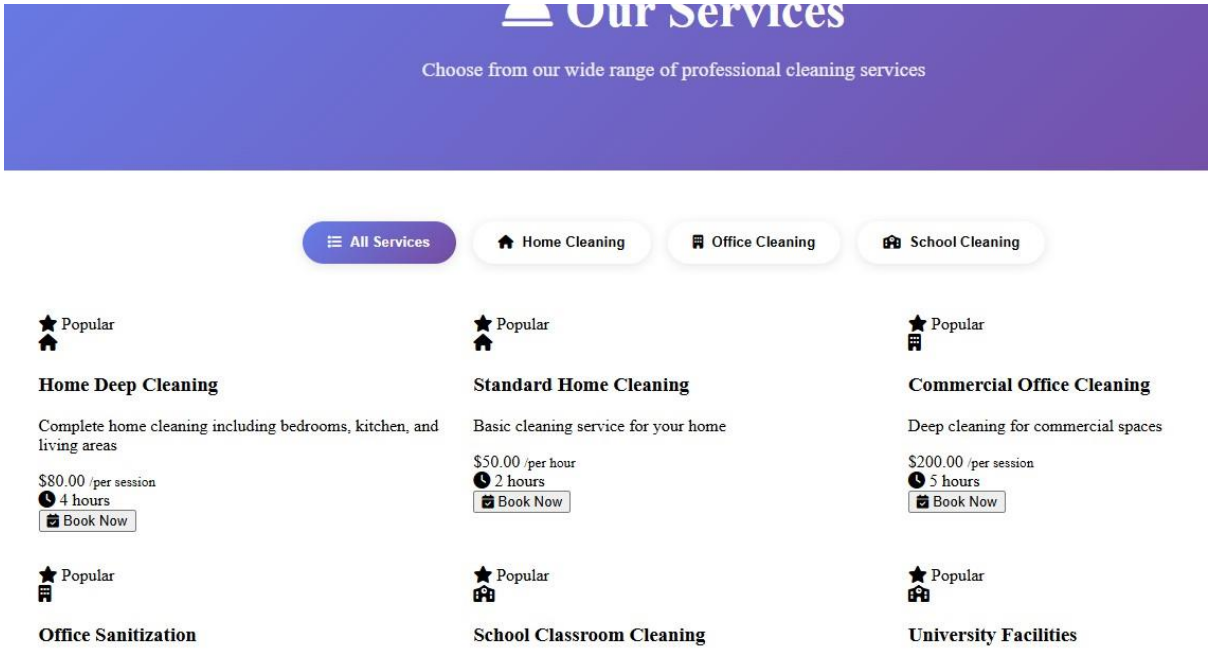
My Jobs

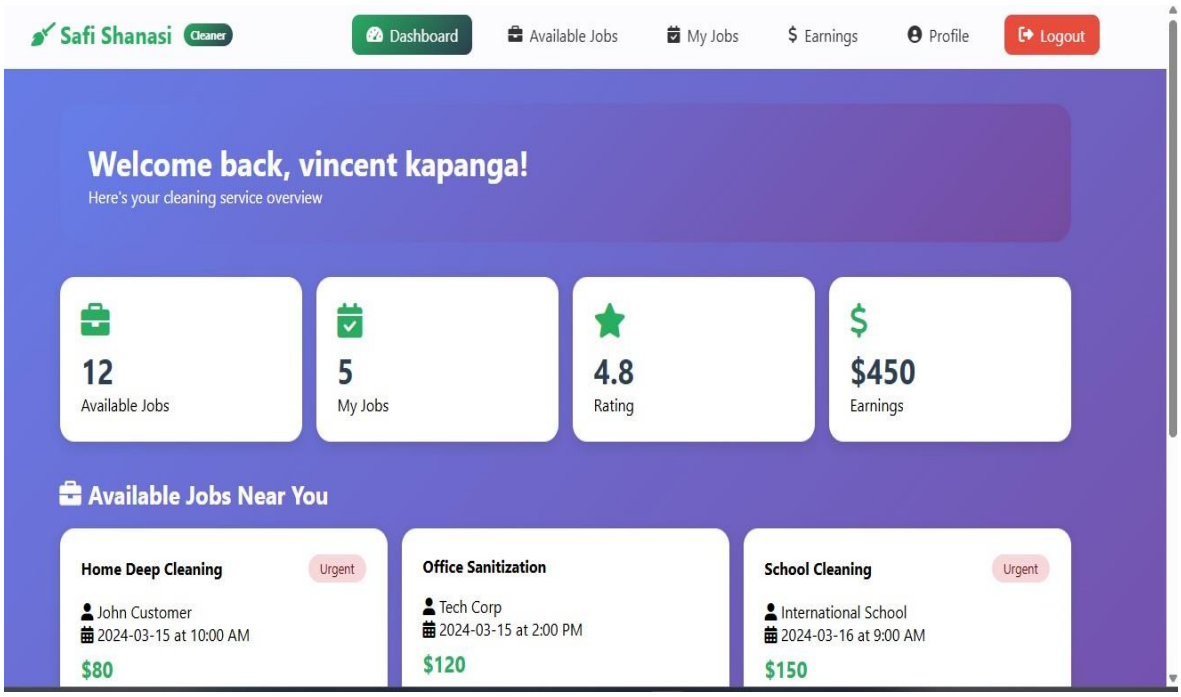
All | Pending | In Progress | Completed

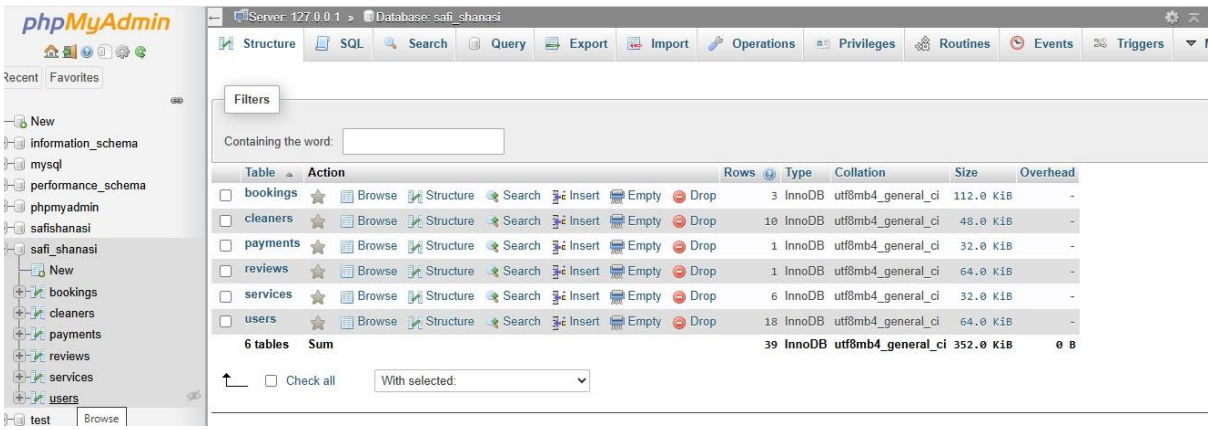
Home Cleaning
John Customer
2024-03-01 10:00 AM $80
Start

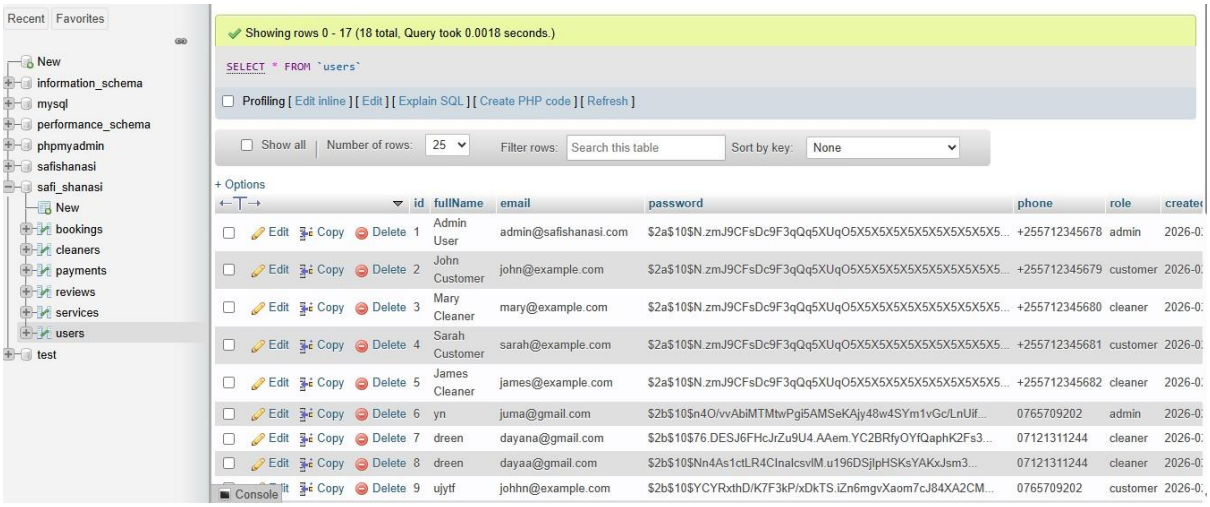
Office Cleaning
Tech Corp
2024-03-01 2:00 PM $120
Complete

School Cleaning


Our Services
Choose from our wide range of professional cleaning services
All Services
Home Cleaning
Office Cleaning
School Cleaning
Popular
Home Deep Cleaning
Complete home cleaning including bedrooms, kitchen, and living areas
$80.00 /per session
4 hours
Book Now
Popular
Standard Home Cleaning
Basic cleaning service for your home
$50.00 /per hour
2 hours
Book Now
Popular
Commercial Office Cleaning
Deep cleaning for commercial spaces
$200.00 /per session
5 hours
Book Now
Popular
Office Sanitization
Popular
School Classroom Cleaning
Popular
University Facilities


Safi Shanasi
Cleaner
Dashboard
Available Jobs
My Jobs
Earnings
Profile
Logout
Welcome back, vincent kapanga!
Here's your cleaning service overview
12
Available Jobs
5
My Jobs
4.8
Rating
$450
Earnings
Available Jobs Near You
Home Deep Cleaning
Urgent
John Customer
2024-03-15 at 10:00 AM
$80
Office Sanitization
Tech Corp
2024-03-15 at 2:00 PM
$120
School Cleaning
Urgent
International School
2024-03-16 at 9:00 AM
$150


phpMyAdmin
Recent
Favorites
New
information_schema
mysql
performance_schema
phpmyadmin
safishanasi
safi_shanasi
New
bookings
cleaners
payments
reviews
services
users
test
Browse
Server: 127.0.0.1 » Database: safi_shanasi
Structure
SQL
Search
Query
Export
Import
Operations
Privileges
Routines
Events
Triggers
Filters
Containing the word:
Table
Action
Rows
Type
Collation
Size
Overhead
bookings 3 InnoDB utf8mb4_general_ci 112.0 KiB -
cleaners 10 InnoDB utf8mb4_general_ci 48.0 KiB -
payments 1 InnoDB utf8mb4_general_ci 32.0 KiB -
reviews 1 InnoDB utf8mb4_general_ci 64.0 KiB -
services 6 InnoDB utf8mb4_general_ci 32.0 KiB -
users 18 InnoDB utf8mb4_general_ci 64.0 KiB -
6 tables Sum 39 InnoDB utf8mb4_general_ci 352.0 KiB 0 B
Check all
With selected:



9. Conclusion

The SafishaNasi App successfully provides an online platform for booking cleaning services. The system integrates frontend, backend, and database technologies to create a functional full stack web application.

10. Future Improvements

Possible improvements include:

- Integration with mobile payment systems such as M-Pesa
- Cleaner rating and review system
- Real-time notifications
- Mobile application version
- Advanced analytics dashboard for administrators